

SYED MUHAMMAD HAMZAH RIZVI

AI Engineer

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PROFESSIONAL SUMMARY

Applied AI engineer with 3+ years building and deploying generative AI and machine learning systems into production. Designed, shipped and scaled GroundedOps, a retrieval-augmented assistant now serving 300+ customer accounts weekly, using hybrid retrieval, cross-encoder reranking and NLI-based grounding checks to suppress hallucinated answers — backed by a 140+ test regression suite and a 34-case evaluation harness gating every change. Separately trained a hierarchical multi-task EfficientNetB0 pipeline on 274k+ facial images to 88.4% coarse-age and 92.6% near-group accuracy. Owns the full lifecycle — pipeline design, evaluation, REST API integration and production debugging — across Python, FastAPI, TensorFlow/Keras and Docker, and communicates clearly with engineering and business stakeholders.

CORE SKILLS

- **Generative AI & LLMs:** RAG systems, AI agents, LLM-powered applications, hybrid retrieval (BM25 + dense embeddings), sigmoid-calibrated cross-encoder reranking, vector search (ChromaDB), prompt design, grounding and hallucination reduction, model evaluation.
- **Machine Learning:** TensorFlow/Keras, EfficientNetB0, VGG16, multi-task deep-learning pipelines, GPU-accelerated training, model evaluation and feature engineering.
- **Deployment & Infrastructure:** Docker (containerised RAG service deployment), REST API design and debugging, FastAPI, webhooks, authentication flows, JSON, Git.
- **Engineering & Collaboration:** Python, C++, test-driven development, production debugging across distributed systems, technical documentation, working with product managers and engineering teams.

PROFESSIONAL EXPERIENCE

Technical Software Consultant — Innovative Technology, Manchester, UK

Mar 2023 – Present

- Own customer deployments of AI-powered biometric and identity-verification products end to end, from architecture discovery through integration, debugging and live production rollout.
- Pair with customer engineering teams to design and build REST API integrations, webhooks and authentication flows connecting AI products to their platforms and external systems.
- Go deep into the stack to unblock rollouts, diagnosing production issues spanning APIs, devices, mobile apps, firmware and networks, feeding fixes back into product and firmware.
- Use AI coding assistants daily to prototype integrations, generate reference implementations and accelerate debugging, validating output against reproduced behaviour rather than accepting it blindly.
- Create reusable technical assets — integration guides, API documentation and diagrams — that make each deployment faster than the last.
- Delivered rollouts across retail, gaming and compliance-focused environments spanning thousands of operational endpoints.

Automation Engineer — Tripack Films Ltd., Pakistan

Mar 2020 – Sep 2021

- Built and maintained industrial automation workflows, integrating control systems and sensors; diagnosed hardware and software issues in real-time operational environments.

Software Consultant — Evolligence, Pakistan

Dec 2019 – Mar 2020

- Built and supported backend application logic using structured APIs and data-driven workflows, collaborating with development teams on delivery.

SELECTED AI PROJECTS

GroundedOps — Production RAG Support Assistant — Python, FastAPI, React, ChromaDB, Ollama, Docker, DeepSeek/OpenAI/Claude APIs

- Deployed to production and now serving 300+ customer accounts weekly through a public embeddable widget and admin console; associated with a 40% reduction in first-line ticket volume and a 55% reduction in resolution time per internal Jira metrics since release (tracked alongside other concurrent initiatives, so not isolated to this tool alone).

- Built a retrieval-augmented QA system with hybrid retrieval (BM25 + dense embeddings via reciprocal rank fusion), sigmoid-calibrated cross-encoder reranking and NLI-based grounding checks to reduce hallucination from local LLMs.
- Containerised the service with Docker for consistent local deployment and reproducible environments across development and testing.
- Diagnosed a query-routing bug by reproducing it in code, tracing it to an unscoped conversation-memory store and a weak follow-up heuristic; rebuilt the path with an LLM Rewrite-Retrieve-Read flow and per-session memory.
- Wrote a pure-stdlib regression suite of 140+ tests covering each diagnosed bug by name, plus an evaluation harness gating every change against 34 real test cases, enabling fast iteration without running model inference on every change.

Age Estimation & Demographic Analysis — *TensorFlow/Keras, EfficientNetB0, VGG16*

- Built a hierarchical multi-task EfficientNetB0 pipeline (gender, coarse age, fine age and continuous age regression) in TensorFlow/Keras on 274k+ facial images from WIKI/IMDB, fine-tuned on UTKFace, with GPU-accelerated training and evaluation tooling.
- Achieved 87.5% gender accuracy, 88.4% coarse-age accuracy, 92.6% near-group (± 1) accuracy and 8.52-year MAE using the hierarchical EfficientNetB0 model, benchmarked against a VGG16 transfer-learning baseline.

EDUCATION

MSc Robotics and Automation (Minor in AI) — University of Salford, UK 2021 – 2022

Graduated with Distinction. Dissertation: facial age and gender estimation using CNNs (TensorFlow/Keras).

BEng Electrical Engineering — NUST, Pakistan 2015 – 2019

Focus: computer vision, machine learning and robotics.